

SOME NON-PRINCIPAL RIGID IDEALS IN GORENSTEIN DOMAINS OF DIMENSION ONE

LARS WINTHER CHRISTENSEN, ALEX GERKO, AND SRIKANTH B. IYENGAR

ABSTRACT. We discuss an example of a rigid non-principal ideal in a one dimensional (commutative) Gorenstein domain, which contradicts a conjecture of C. Huneke and R. Wiegand. The construction and its analysis were discovered by Codex, when prompted by one of the authors to verify the conjecture or find a counterexample. A proof of the conjecture, also discovered by Codex, is presented when the ring is equicharacteristic and its embedding dimension is at most three.

1. INTRODUCTION

A conjecture of Huneke and Wiegand [8, pp. 473] postulates that any ideal I in a Gorenstein (commutative, noetherian) local domain R with the property that the R -module $I \otimes_R \operatorname{Hom}_R(I, R)$ is torsion-free, must be free. The condition the tensor product is torsion-free is equivalent to the condition that

$$\operatorname{Ext}_R^1(I, I) = 0,$$

that is to say, that the ideal I is *rigid*; see, for instance, [7, Proposition 4.3]. This conjecture has been verified in various special cases; see, for instance, [7].

In this note we describe a general method, discovered by ChatGPT, for constructing counter-examples to this conjecture; see Propositions 2.2 and 3.2. A concrete example realizing this method is given in Example 3.3. In fact, in this example the ring R is standard-graded, and the ideal I is two-generated, by linear forms; we refer to Example 3.3 for other noteworthy features.

At about the same time as these examples were discovered, Son Pham [11] found another example, using GPT-5.6 Pro; this has been verified independently by Huneke [11]. This example is a semi-group ring, and hence also a graded ring, but it is not standard-graded.

How this was found. Sol Ultra on Codex, prompted to prove the Huneke–Wiegand conjecture for Gorenstein rings or find counterexample, and separately prompted to either prove it for complete intersections or find counterexample. The manuscript is a streamlined version of text generated by Codex.

Acknowledgements. The authors express their thanks to Craig Huneke for sharing his verification of Song Pham’s example.

Date: 21 August 2026.

2020 Mathematics Subject Classification. 13C13 (primary); 13D07 (secondary).

Key words and phrases. Torsion in tensor products, Gorenstein ring, rigid ideal.

2. GORENSTEIN DOMAINS OF DIMENSION ONE

Given a field K and a symmetric bilinear form on a finite dimensional K -vector space V , one has a finite dimensional commutative graded K -algebra $K \oplus V \oplus K$, where the only non-trivial product $V \times V \rightarrow K$ is given by the form. Moreover, this algebra is Gorenstein if and only if the form is not degenerate; see, for instance, [9, (3.15)].

This section gives a method for constructing Gorenstein domains of Krull dimension one, allegedly inspired by the work of Eisenbud and Popescu [5, Theorem 7.1], and also Stevens [12].

Definition 2.1. Let K be a field and $K \subset L$ a finite extension, such that its degree $n := [L : K]$ is even. Fix a K -subspace $H \subset L$ containing K and of index 1, that is to say, $\text{rank}_K H = n - 1$.

Let W be a K -linear subspace of H satisfying the following conditions:

$$K \subsetneq W, \quad \text{rank}_K W = \frac{n}{2}, \quad \text{and} \quad W^2 \subseteq H.$$

Let t be an indeterminate and $L[t]$ the standard graded L -algebra, so $|t| = 1$. Since $W^2 \subseteq H$ holds, the graded K -vector subspace of $L[t]$ given by

$$R := K \oplus Wt \oplus Ht^2 \bigoplus_{i \geq 3} Lt^i$$

is a K -subalgebra. Here are the principal features of this ring.

Proposition 2.2. *The ring R is graded Gorenstein (in particular, noetherian) domain of Krull dimension one. As a K -algebra it is generated in degrees one and two; if $W^2 = H$, then it is standard graded.*

Proof. Clearly, one has inclusion of subrings $K[t] \subset R \subset L[t]$. Since L/K is finite dimensional, $L[t]$ is finitely generated, and even free, as an $K[t]$ -module. Thus, its $K[t]$ -submodule R is also finitely generated, and free. This justifies the claim that the ring R is a noetherian domain and of Krull dimension one.

In the remainder of the proof, it is useful to consider the K -linear functional $f: L \rightarrow L/H \cong K$, and the symmetric K -bilinear form

$$\langle -, - \rangle: L \times L \longrightarrow K \quad \text{where } \langle x, y \rangle = f(xy).$$

This form is nondegenerate: Given $0 \neq x \in L$ one has $\langle x, x^{-1}z \rangle \neq 0$ for $z \in L \setminus H$.

For any K -subspace $U \subseteq L$ one has the orthogonal subspace

$$(2.1) \quad U^\perp := \{x \in L \mid \langle x, - \rangle = 0 \text{ on } U\}.$$

This subspace satisfies $\text{rank}_K U + \text{rank}_K U^\perp = \text{rank}_K L$. We claim

$$W^\perp = W \quad \text{and} \quad H^\perp = K.$$

Indeed, the hypothesis that $W^2 \subseteq H$ is equivalent to $W \subseteq W^\perp$; equality holds because of the hypothesis that $\text{rank}_K W = n/2$ and (2.1), which justifies the equality on the left. As to the one on the right, again from (2.1) we deduce that $\text{rank}_K H^\perp = 1$. Since $K \subset H = \text{Ker } f$, we conclude that $H^\perp = K$.

Since $H^\perp = K$, for any element w in $W \setminus K$ one has that $wH \not\subseteq H$. Thus as a K -algebra R is generated by its elements of degree one and two; evidently, it is standard graded if also $W^2 = H$.

It remains to verify that R is Gorenstein. Since R is a domain any nonzero element, in particular t , is not a zero divisor on it. A direct calculation yields an isomorphism of K -algebras

$$R/Rt \cong K \oplus (W/K)t \oplus (H/W)t^2 \oplus (L/H)t^3.$$

The elements of degree three are in the socle. We verify that these are the only ones; since $\text{rank}_K(L/H) = 1$ this is equivalent to R/Rt Gorenstein; see [2, 3.2.10]

Suppose the residue class of wt is in the socle of R/Rt . Then, from the structure of R , we get $wH \subseteq H$, that is to say w in $H^\perp = K$. Then $wt = 0$ in R/Rt . Similarly if the residue class of ht^2 is in the socle, then $hW \subseteq H$, so h is in $W^\perp = W$, and hence $ht^2 = 0$ in R/Rt .

This completes the verification that R/Rt is Gorenstein. Since t is a homogeneous element and not a zero-divisor, we conclude that R is Gorenstein; see [2, 3.1.19]. $\square$

3. RIGID IDEALS

In this section, we identify some conditions on the rings described in the previous section that ensure that they contain rigid ideals that are not principal.

In the setup of Definition 2.1, fix an element α in $W \setminus K$ and consider the ideal

$$I := (t, \alpha t)R$$

of R . This ideal is clearly not principal, and hence not free. The claim is that, for a suitable choice of subspace W , the ideal I is rigid. Not every choice of W works, as can be verified by a direct check.

We focus now on a special situation when it does, leaving the general analysis that leads to this choice to the end of this section.

Definition 3.1. Let $L = K[\alpha]$, where α has degree $n \equiv 0 \pmod{4}$, with $\alpha \notin K$, so $n \geq 4$, and $\alpha^n \in K^\times$. Thus $1, \alpha, \alpha^2, \dots, \alpha^{n-1}$ is a basis for L over K . For $i \geq 0$, let L_i denote the K -subspace of L spanned by $1, \dots, \alpha^i$. Thus one has a filtration of L by K -subspaces

$$K = L_0 \subset L_1 \subset \dots \subset L_{n-1} = L.$$

Choose a K -subspace $V \subset L$ containing $L_0 = K$ and such that the following conditions hold:

- (1) $\text{rank}_K V = n/4$;
- (2) the sum $V + \alpha V + \alpha^2 V$ is direct;
- (3) $V^2 = L_{n-4}$;
- (4) $L = VL_{n-3}$.

With V as above, set $W := V \oplus \alpha V$ and $H := L_{n-2}$. Recall that

$$R := K \oplus Wt \oplus Ht^2 \bigoplus_{i \geq 3} Lt^i$$

viewed as a graded K -subalgebra of $L[t]$.

Proposition 3.2. *In the setup of Definition 3.1, the ring R is a standard graded Gorenstein domain of Krull dimension one. The ideal $(t, \alpha t)$ is rigid, but not free.*

Proof. The claims about the ring R follows from Proposition 2.2; observe that $W^2 = H$, by condition (3) in Definition 3.1.

To verify that $(t, \alpha t)$ is rigid, we use the criterion [7, Corollary 4.9]; that statement is for local rings, but the argument applies equally well for homogeneous ideals in graded rings. Thus, we have to verify the equality:

$$((t) :_R \alpha t) \cap ((\alpha t) :_R t) = ((t) :_R \alpha t)((\alpha t) :_R t).$$

To begin with, observe that

$$((\alpha t) :_R t) = \alpha \cdot ((t) :_R \alpha t);$$

keep in mind that α itself is not an element of R . Thus the desired equality is that

$$J \cap \alpha J = \alpha J^2 \quad \text{where } J = ((t) :_R \alpha t).$$

Since $(t) = Kt \oplus Wt^2 \oplus Ht^3 \oplus_{i \geq 4} Lt^i$, using condition (2) in Definition 3.1 a direct verification yields

$$J = Vt \oplus L_{n-3}t^2 \bigoplus_{i \geq 3} Lt^i \quad \text{and hence} \quad \alpha J = \alpha Vt \oplus \alpha L_{n-3}t^2 \bigoplus_{i \geq 3} Lt^i.$$

Observing that $V \cap \alpha V = 0$, by condition (2) in Definition 3.1, one thus gets that

$$J \cap \alpha J = (L_{n-3} \cap \alpha L_{n-3})t^2 \bigoplus_{i \geq 3} Lt^i.$$

On the other hand, a direct computation yields the first equality below

$$\begin{aligned} \alpha J^2 &= \alpha(V^2t^2 \oplus VL_{n-3}t^3 \bigoplus_{i \geq 4} Lt^i) \\ &= \alpha(L_{n-4}t^2 \oplus Lt^3 \bigoplus_{i \geq 4} Lt^i) \\ &= \alpha L_{n-4}t^2 \oplus \bigoplus_{i \geq 3} Lt^i. \end{aligned}$$

The second equality holds by conditions (3) and (4) in Definition 3.1, and the third follows from $\alpha L = L$. Thus, to verify that $J \cap \alpha J = \alpha J^2$, it remains to verify that

$$L_{n-3} \cap \alpha L_{n-3} = \alpha L_{n-4}.$$

This is clear, given the description of the L_i . □

A numerical constraint. In the notation above, suppose $d := \text{rank}_K V$, so that $\text{rank}_K W = 2d$ and $\text{rank}_K L = 4d$. Thus the condition that $V^2 = L_{n-4}$, from Definition 3.1(2), implies that

$$\binom{d+1}{2} = \text{rank}_K \text{Sym}_K^2(V) \geq \text{rank}_K L_{n-4} = 4d - 3.$$

This leads to the condition that

$$(d-1)(d-6) \geq 0.$$

One cannot have $d = 1$, for then $V = K$ so $VL_{n-3} = L_{n-3} \neq L$. Thus the smallest possible value for d that would permit V to satisfy the constraints in Definition 3.1 is 6. Then $\text{rank}_K L = 24$.

When $d = 6$, one has

$$\text{rank}_K \text{Sym}_K^2(V) = 21 = \text{rank}_K L_{20}$$

so condition (2) in Definition 3.1 is equivalent to the non-vanishing of a determinant of size 21×21 .

These considerations motivate the following example.

Example 3.3. Let $K = \mathbb{Q}$ and $L := \mathbb{Q}[\alpha]/(\alpha^{24} - 2)$. The polynomial $x^{24} - 2$ is irreducible, by, say the Eisenstein criterion, so $\text{rank}_K L = 24$. Let V be the $\mathbb{Q}$ -subspace of L spanned by the six elements

$$\begin{aligned} &1, \quad \alpha - \alpha^{15}, \quad \alpha^2 - \alpha^{12} - \alpha^{16} \\ &\alpha^4 - \alpha^{14}, \quad \alpha^6 - \alpha^{13} + \alpha^{20}, \quad \alpha^9 + \alpha^{16} + \alpha^{19}. \end{aligned}$$

One can verify directly by hand, or using a computer algebra system, that V satisfies the conditions listed in Definition 3.1; see Section 4 for a Macaulay 2 verification. Thus Propositions 2.2 and 3.2 justify the following claims.

With V and L as above, and $W := V \oplus \alpha V$, the subring

$$R := \mathbb{Q} \oplus Wt \oplus L_{22}t^2 \oplus_{i \geq 3} Lt^i$$

of $L[t]$ is a standard graded Gorenstein $\mathbb{Q}$ -algebra and a domain. Moreover, the ideal $I := (t, \alpha t)$ is rigid, but not principal.

The ideal I has the added property that the evaluation map

$$I \otimes_R \text{Hom}_R(I, R) \longrightarrow R$$

is an isomorphism onto $\mathfrak{m}$, the homogeneous maximal ideal of R .

Remark 3.4. One can localize at the homogeneous maximal ideal $\mathfrak{m} := R^{\geq 1}$ and get an example of local Gorenstein domain of Krull dimension one that provides a counterexample to the conjecture of Huneke and Wiegand.

A commutative algebra analogue of a conjecture of Auslander and Reiten [1] states that over any Gorenstein ring R , if a finitely generated R -module M satisfies $\text{Ext}_R^i(M, M) = 0$ for $i \geq 1$, then M is free. For the ring R and ideal I in Example 3.3 one can verify that

$$\text{Ext}_R^2(I, I) \neq 0.$$

Thus I is not a counterexample to the Auslander-Reiten conjecture.

An ansatz leading to Definition 3.1. Let R be a ring as in Definition 2.1 and set $I = (t, \alpha t)$, for some element α in $L \setminus K$. As explained in the proof of Proposition 3.2, the ideal I is rigid precisely when there is an equality

$$(3.1) \quad J \cap \alpha J = \alpha J^2, \quad \text{where } J := ((t) :_R \alpha t).$$

Consider the following K -subspaces

$$V := \alpha^{-1}W \cap W \quad \text{and} \quad E := \alpha^{-1}H \cap H.$$

It is straightforward to verify that

$$J = Vt \oplus Et^2 \oplus_{i \geq 3} Lt^i,$$

and hence condition (3.1) translates to

$$(3.2) \quad V \cap \alpha V = (0), \quad E \cap \alpha E = \alpha V^2, \quad \text{and} \quad L = VE.$$

Keep in mind that for any K -subspace U of L , one has $L = \alpha U$ if and only if $L = U$. The first condition means that $V \oplus \alpha V \subseteq W$ so a first guess is to take

$$W := V \oplus \alpha V.$$

Since $\text{rank}_K W$ should be $n/2$ for Proposition 2.2 to apply, $\text{rank}_K V = n/4$ must hold, which explains condition (1) in Definition 3.1.

Then since $V = \alpha^{-1}W \cap W$, it follows that the sum

$$V + \alpha V + \alpha^2 V$$

is direct, which explains condition (2) in Definition 3.1.

Now suppose that $L = K[\alpha]$ is as in Definition 3.1. In the notation introduced there, one has $H = L_{n-2}$, so

$$E = \alpha^{-1}H \cap H = L_{n-3}.$$

Thus the two conditions on the right in (3.2) translate to conditions (3) and (4) in Definition 3.1.

4. VERIFICATION WITH MACAULAY2

In this section we provide *Macaulay2* code that we used to verify some of the assertions about Example 3.3.

Example 3.3 revisited. The *Macaulay2* [6] script below verifies that the subspace V in 3.3 satisfies the conditions in Definition 3.1. It is available from [3].

```
-- Setting up K and L

K = QQ; L = K[a]/ideal(a^24-2);

-- Defining a function to compute the standard coordinates
-- of a polynomial in L

polyToList = p -> ( apply(24, i -> coefficient(a^i,p)) );

-- Setting up the subspace V of L

gensV = {1_L, a - a^15, a^2 - a^12 - a^16, a^4 - a^14,
         a^6 - a^13 + a^20, a^9 + a^16 + a^19};

V = image transpose matrix apply( gensV, b -> polyToList(b) );
rank V -- rank is 6

-- Setting up alpha times V and alpha^2 times V

gensaV = apply( gensV, b-> a*b );
aV = image transpose matrix apply( gensaV, b -> polyToList(b) );
rank aV -- output: 6

gensa2V = apply( gensV, b-> a^2*b );
a2V = image transpose matrix apply( gensa2V, b -> polyToList(b) );
rank a2V -- output: 6

-- Verifying condition 3.1(1)

rank(V+aV+a2V) == rank V + rank aV + rank a2V -- output: true

-- Verifying condition 3.1(2)
```

```
gensV2 = flatten apply( 6,i -> apply( i+1, j -> (gensV#i)*(gensV#j) ));
V2 = image transpose matrix apply( gensV2, b -> polyToList(b) );
rank V2 -- output: 21
```

```
basisL20 = apply( 21, i -> a^i );
L20 = image transpose matrix apply( basisL20, b -> polyToList(b) );
V2 == L20 -- output: true
```

```
-- Verifying condition 3.1(3)
```

```
gensVL21 = flatten apply( gensV, b -> apply( 22, i -> a^i*b ));
VL21 = image transpose matrix apply( gensVL21, b -> polyToList(b) );
rank VL21 -- output: 24, so it is all of L
```

Next we realize the ring from Example 3.3 as a quotient of a polynomial algebra. Set $Q = \mathbb{Q}[x_1, \dots, x_6, y_1, \dots, y_6]$ and consider the canonical ring homomorphism $\phi: Q \rightarrow L[t]$ that maps the indeterminates to the basis for $Wt = (V \oplus \alpha V)t$ so that

$$\begin{aligned} x_1 &\mapsto t, \dots, x_6 \mapsto (\alpha^9 + \alpha^{16} + \alpha^{19}) \\ y_1 &\mapsto \alpha t, \dots, y_6 \mapsto \alpha(\alpha^9 + \alpha^{16} + \alpha^{19})t. \end{aligned}$$

By construction, the image of ϕ in degree 1 is R_1 , and since R and Q are standard graded, the kernel of ϕ is the defining ideal of R as a quotient of Q . Recalling that the Hilbert series of R is $1 + 12t + 23t^2 + \sum_{i=3}^{\infty} 24t^i$ and $\text{rank}_K \text{Sym}^2(Q_1) = \binom{13}{2} = 78$, one sees that the kernel of ϕ is generated by $78 - 23 = 55$ quartics.

The script

```
-- Setting up the polynomial algebra

Q = QQ[x_1..x_6,y_1..y_6];

-- Setting up the basis for Wt = (V + aV)t

Lt = L[t];

gensWt = apply( gensV|gensaV, b -> b*t );

-- Define the canonical map phi: Q -> L[t] that maps the
-- 12 indeterminates to the basis for Wt

phi = map(Lt,Q,gensWt);

I = ker phi;

-- Set up the 78 - 23 = 55 relations in Q to define R

rels = { x_2^2 - y_1^2 - 2*y_1*y_4,
  x_2*x_6 + 2*x_1^2 - 36*x_1*x_2 + 72*x_1*x_3 + 42*x_1*x_4 - 2*x_1*x_5
+ 36*x_1*y_1 - 74*x_1*y_2 + 4*x_1*y_3 + 72*x_1*y_4 + 12*x_1*y_5 +
x_1*y_6 + 2*y_1^2 + 2*y_1*y_2 - 48*y_1*y_3 + 2*y_1*y_4 - 12*y_1*y_6
- 2*x_2*x_3 - 6*x_2*x_5 - 36*x_2*y_5 + 18*y_2^2,
  x_2*y_1 - x_1*y_2,
  x_2*y_2 + 24*x_1*x_2 - 48*x_1*x_3 - 28*x_1*x_4 - 24*x_1*y_1 +
```

$$\begin{aligned}
& 48x_1y_2 - x_1y_3 - 48x_1y_4 - 8x_1y_5 - 3y_1y_2 + \\
& 32y_1y_3 + 8y_1y_6 + x_2x_3 + 4x_2x_5 + 24x_2y_5 - \\
& 12y_2^2, \\
& x_2y_3 + 2x_1x_2 - 4x_1x_3 - 3x_1x_4 - 2x_1y_1 + 4x_1y_2 \\
& - 4x_1y_4 + 2y_1y_3 + 2x_2y_5 - 2y_2^2, \\
& x_2y_4 + x_1x_5 + 3x_1y_2 - x_1y_3 - 3y_1^2 + y_1y_2 - \\
& 4y_1y_4, \\
& x_2y_6 - 6x_1x_2 + 12x_1x_3 + 10x_1x_4 + 8x_1y_1 - \\
& 12x_1y_2 + 12x_1y_4 + 2x_1y_5 - 10y_1y_3 - 3y_1y_6 - \\
& 2x_2x_5 - 6x_2y_5 + 4y_2^2, \\
& x_3^2 - 2x_1^2 + 4x_1x_2 - 8x_1x_3 - 6x_1x_4 - 4x_1y_1 + \\
& 8x_1y_2 - 8x_1y_4 + 4y_1y_3 + 4x_2y_5 - 3y_2^2, \\
& x_3x_4 + x_1x_5 + 2x_1y_2 - x_1y_3 - 4y_1^2 + y_1y_2 - \\
& 4y_1y_4, \\
& x_3x_5 + 2x_1x_2 - 4x_1x_3 + x_1x_4 - 4x_1y_1 + 4x_1y_2 - \\
& 4x_1y_4 - y_1y_3 + x_2y_5, \\
& x_3x_6 - 18x_1x_2 + 36x_1x_3 + 22x_1x_4 + 20x_1y_1 - \\
& 36x_1y_2 + 36x_1y_4 + 5x_1y_5 - 24y_1y_3 - 5y_1y_6 - \\
& 3x_2x_5 - 18x_2y_5 + 10y_2^2, \\
& x_3y_1 - x_1y_3, \\
& x_3y_2 + 2x_1x_2 - 4x_1x_3 - 3x_1x_4 - 2x_1y_1 + 4x_1y_2 \\
& - 4x_1y_4 + 2y_1y_3 + 2x_2y_5 - 2y_2^2, \\
& x_3y_3 - 5x_1x_2 - 2x_1x_6 + 3x_1y_1 - 2x_1y_2 + 7x_1y_4 \\
& + 2y_1^2 - 4x_2x_4, \\
& x_3y_4 + 22x_1x_2 - 44x_1x_3 - 27x_1x_4 - 22x_1y_1 + \\
& 44x_1y_2 - 44x_1y_4 - 7x_1y_5 - 2y_1y_2 + 30y_1y_3 + \\
& 8y_1y_6 + 4x_2x_5 + 22x_2y_5 - 11y_2^2, \\
& x_3y_5 + x_1x_6 + 2x_1y_2 - 2x_1y_3 - 4y_1^2 + 2y_1y_2 - \\
& 2y_1y_4, \\
& x_3y_6 + 6x_1x_2 - 5x_1x_3 + 2x_1x_6 - 6x_1y_1 + 7x_1y_2 \\
& - 11x_1y_4 + 3x_2x_4 + 2x_2y_5, \\
& x_4^2 + 3x_1x_2 - 6x_1x_3 - 2x_1x_4 - 3x_1y_1 + 6x_1y_2 - \\
& 6x_1y_4 + 2y_1y_3 + 3x_2y_5 - 2y_2^2, \\
& x_4x_5 - 2x_1^2 - 24x_1x_2 + 48x_1x_3 + 28x_1x_4 + \\
& 24x_1y_1 - 48x_1y_2 + 2x_1y_3 + 48x_1y_4 + 8x_1y_5 + \\
& x_1y_6 + 2y_1y_2 - 32y_1y_3 - 8y_1y_6 - 2x_2x_3 - 4x_2x_5 \\
& - 24x_2y_5 + 12y_2^2, \\
& x_4x_6 + 4x_1x_2 - x_1x_5 + 2x_1x_6 - 4x_1y_1 - x_1y_2 + \\
& 2x_1y_3 - 6x_1y_4 + y_1^2 - 2y_1y_2 + 3y_1y_4 + 2x_2x_4, \\
& x_4y_1 - x_1y_4, \\
& x_4y_2 + x_1x_5 + 3x_1y_2 - x_1y_3 - 3y_1^2 + y_1y_2 - \\
& 4y_1y_4, \\
& x_4y_3 + 22x_1x_2 - 44x_1x_3 - 27x_1x_4 - 22x_1y_1 + \\
& 44x_1y_2 - 44x_1y_4 - 7x_1y_5 - 2y_1y_2 + 30y_1y_3 + \\
& 8y_1y_6 + 4x_2x_5 + 22x_2y_5 - 11y_2^2, \\
& x_4y_4 - 4x_1x_2 - x_1x_6 + 4x_1y_1 - x_1y_2 + 7x_1y_4 + \\
& y_1^2 - 3x_2x_4, \\
& x_4y_5 + 4x_1x_2 - 8x_1x_3 - 4x_1x_4 - 6x_1y_1 + 8x_1y_2 \\
& - 8x_1y_4 + 4y_1y_3 + y_1y_6 + 4x_2y_5 - 2y_2^2, \\
& x_4y_6 - 38x_1x_2 + 76x_1x_3 + 47x_1x_4 - 2x_1x_5 + \\
& 38x_1y_1 - 78x_1y_2 + 4x_1y_3 + 76x_1y_4 + 13x_1y_5 + \\
& 2x_1y_6 + 2y_1^2 + 2y_1y_2 - 52y_1y_3 + 2y_1y_4 - \\
& 14y_1y_6 - 2x_2x_3 - 7x_2x_5 - 38x_2y_5 + 19y_2^2,
\end{aligned}$$

$$\begin{aligned}
& x_5^2 + 4x_1x_2 + x_1x_3 + 4x_1x_6 - 4x_1y_1 + 5x_1y_2 - \\
& 6x_1y_4 - 12y_1^2 + 2x_2x_4, \\
& x_5x_6 + 5x_1x_2 - 2x_1x_3 - 7x_1y_1 + 2x_1y_2 - 4x_1y_4 \\
& + 2x_2y_5, \\
& x_5y_1 - x_1y_5, \\
& x_5y_2 - x_2y_5, \\
& x_5y_3 + x_1x_6 + 2x_1y_2 - 2x_1y_3 - 4y_1^2 + 2y_1y_2 - \\
& 2y_1y_4, \\
& x_5y_4 + 4x_1x_2 - 8x_1x_3 - 4x_1x_4 - 6x_1y_1 + 8x_1y_2 \\
& - 8x_1y_4 + 4y_1y_3 + y_1y_6 + 4x_2y_5 - 2y_2^2, \\
& x_5y_5 - 144x_1x_2 + 288x_1x_3 + 168x_1x_4 - 2x_1x_5 + \\
& 144x_1y_1 - 290x_1y_2 + 15x_1y_3 + 288x_1y_4 + 48x_1y_5 + \\
& 4x_1y_6 + 2y_1^2 + 15y_1y_2 - 192y_1y_3 + 2y_1y_4 - \\
& 48y_1y_6 - 12x_2x_3 - 24x_2x_5 - 144x_2y_5 + 72y_2^2, \\
& x_5y_6 + 4x_1x_2 + 2x_1x_6 - 4x_1y_1 + 5x_1y_2 + 2x_1y_3 \\
& - 6x_1y_4 - 7y_1^2 - 2y_1y_2 + 2x_2x_4, \\
& x_6^2 + 26x_1x_2 - 52x_1x_3 - 31x_1x_4 - 30x_1y_1 + \\
& 52x_1y_2 - 52x_1y_4 - 4x_1y_5 + 33y_1y_3 + 4y_1y_6 + \\
& 4x_2x_5 + 26x_2y_5 - 14y_2^2, \\
& x_6y_1 - x_1y_6, \\
& x_6y_2 - 6x_1x_2 + 12x_1x_3 + 10x_1x_4 + 8x_1y_1 - \\
& 12x_1y_2 + 12x_1y_4 + 2x_1y_5 - 10y_1y_3 - 3y_1y_6 - \\
& 2x_2x_5 - 6x_2y_5 + 4y_2^2, \\
& x_6y_3 + 6x_1x_2 - 5x_1x_3 + 2x_1x_6 - 6x_1y_1 + 7x_1y_2 \\
& - 11x_1y_4 + 3x_2x_4 + 2x_2y_5, \\
& x_6y_4 - 38x_1x_2 + 76x_1x_3 + 47x_1x_4 - 2x_1x_5 + \\
& 38x_1y_1 - 78x_1y_2 + 4x_1y_3 + 76x_1y_4 + 13x_1y_5 + \\
& 2x_1y_6 + 2y_1^2 + 2y_1y_2 - 52y_1y_3 + 2y_1y_4 - \\
& 14y_1y_6 - 2x_2x_3 - 7x_2x_5 - 38x_2y_5 + 19y_2^2, \\
& x_6y_5 + 4x_1x_2 + 2x_1x_6 - 4x_1y_1 + 5x_1y_2 + 2x_1y_3 \\
& - 6x_1y_4 - 7y_1^2 - 2y_1y_2 + 2x_2x_4, \\
& x_6y_6 + 4x_1x_2 + 4x_1x_3 - 2x_1x_6 - 4x_1y_1 - 6x_1y_2 \\
& - x_1y_4 - 2y_1^2 - x_2x_4, \\
& y_2y_3 - 6x_1x_2 - 2x_1x_6 + 6x_1y_1 - 2x_1y_2 + 9x_1y_4 \\
& + 2y_1^2 - 4x_2x_4, \\
& y_2y_4 + 34x_1x_2 - 68x_1x_3 - 41x_1x_4 - 34x_1y_1 + \\
& 68x_1y_2 - x_1y_3 - 68x_1y_4 - 11x_1y_5 - 2y_1y_2 + \\
& 46y_1y_3 + 12y_1y_6 + x_2x_3 + 6x_2x_5 + 34x_2y_5 - \\
& 17y_2^2, \\
& y_2y_5 - 2x_1x_2 - x_1x_6 + 2x_1y_1 - 2x_1y_3 + 3x_1y_4 + \\
& 2y_1y_2 - 2y_1y_4 - x_2x_4, \\
& y_2y_6 + 8x_1x_2 - 3x_1x_3 + 2x_1x_6 - 8x_1y_1 + 5x_1y_2 \\
& - 11x_1y_4 - y_1y_5 + 3x_2x_4 + x_2y_5, \\
& y_3^2 + 24x_1x_2 - 48x_1x_3 - 28x_1x_4 + 4x_1x_5 - \\
& 24x_1y_1 + 55x_1y_2 - 6x_1y_3 - 48x_1y_4 - 8x_1y_5 - \\
& 2x_1y_6 - 9y_1^2 + 32y_1y_3 - 9y_1y_4 + 8y_1y_6 + 2x_2x_3 \\
& + 4x_2x_5 + 24x_2y_5 - 12y_2^2, \\
& y_3y_4 - x_1x_2 - 2x_1x_4 + x_1y_1 + 2y_1y_3 + y_1y_5 - \\
& 2y_2^2, \\
& y_3y_5 - 16x_1x_2 + 32x_1x_3 + 16x_1x_4 + 16x_1y_1 - \\
& 32x_1y_2 + 2x_1y_3 + 32x_1y_4 + 4x_1y_5 + x_1y_6 + \\
& 2y_1y_2 - 20y_1y_3 - 4y_1y_6 - 2x_2x_3 - 2x_2x_5 - \\
& 16x_2y_5 + 8y_2^2,
\end{aligned}$$

```

y_3*y_6 + 4*x_1*x_2 - 3*x_1*x_5 + 2*x_1*x_6 - 4*x_1*y_1 - 3*x_1*y_2
+ 2*x_1*y_3 - 6*x_1*y_4 + 2*x_1*y_6 + 3*y_1^2 + 5*y_1*y_4 +
2*x_2*x_4,
y_4^2 + 12*x_1*x_2 - 24*x_1*x_3 - 14*x_1*x_4 + 3*x_1*x_5 -
12*x_1*y_1 + 29*x_1*y_2 - 4*x_1*y_3 - 24*x_1*y_4 - 4*x_1*y_5 -
x_1*y_6 - 5*y_1^2 + y_1*y_2 + 16*y_1*y_3 - 5*y_1*y_4 + 4*y_1*y_6 +
x_2*x_3 + 2*x_2*x_5 + 12*x_2*y_5 - 6*y_2^2,
y_4*y_5 - 2*x_1*x_2 + x_1*x_3 + 2*x_1*y_1 - x_1*y_2 + 3*x_1*y_4 -
2*y_1^2 + y_1*y_5 - x_2*x_4 - x_2*y_5,
y_4*y_6 - 3*x_1*x_2 + 10*x_1*x_3 + 10*x_1*x_4 + 3*x_1*y_1 -
10*x_1*y_2 + 10*x_1*y_4 + 2*x_1*y_5 - 10*y_1*y_3 - y_1*y_5 -
2*y_1*y_6 - 2*x_2*x_5 - 5*x_2*y_5 + 4*y_2^2,
y_5^2 + 6*x_1*x_2 - 12*x_1*x_3 - 7*x_1*x_4 - 6*x_1*y_1 + 12*x_1*y_2
- 12*x_1*y_4 + 2*x_1*y_5 + 8*y_1*y_3 - 2*x_2*x_5 + 6*x_2*y_5 -
3*y_2^2,
y_5*y_6 - 80*x_1*x_2 + 160*x_1*x_3 + 96*x_1*x_4 - 2*x_1*x_5 +
80*x_1*y_1 - 162*x_1*y_2 + 9*x_1*y_3 + 160*x_1*y_4 + 28*x_1*y_5 +
2*x_1*y_6 + 2*y_1^2 + 10*y_1*y_2 - 108*y_1*y_3 + 2*y_1*y_4 -
28*y_1*y_6 - 7*x_2*x_3 - 14*x_2*x_5 - 80*x_2*y_5 + 40*y_2^2,
y_6^2 - 24*x_1*x_2 + 48*x_1*x_3 + 28*x_1*x_4 + x_1*x_5 + 24*x_1*y_1
- 41*x_1*y_2 + 5*x_1*y_3 + 48*x_1*y_4 + 8*x_1*y_5 - 2*x_1*y_6 -
7*y_1^2 - 3*y_1*y_2 - 32*y_1*y_3 - 5*y_1*y_4 - 8*y_1*y_6 - 2*x_2*x_3
- 4*x_2*x_5 - 24*x_2*y_5 + 12*y_2^2 };

-- Verify the number of relations

#rels -- output: 55

-- Verify that the relations generate the kernel of phi

ideal rels == ker phi -- output: true

R = Q/(ideal rels)

-- Verify that the Hilbert function of R is as expected

apply(6,d -> hilbertFunction(d,R)) -- output: {1, 23, 23, 24, 24, 24}

-- Set up the ideals in R corresponding to t and at

X = ideal(x_1);
Y = ideal(y_1);

-- Verify the condition on these ideals from Huneke-Iyengar-Wiegand

(X:Y)*(Y:X) == intersect(X:Y,Y:X) -- output: true

-- Notice the relation between the colon ideals observed in proof of 3.2

mingens (X:Y) -- output: | x_6 x_5 x_4 x_3 x_2 x_1 |
mingens (Y:X) -- output: | y_6 y_5 y_4 y_3 y_2 y_1 |

```

5. CODIMENSION TWO

In this section we verify that conjecture of Huneke and Wiegand holds when, in addition, the Gorenstein ring R is of codimension at most 2; see Proposition 5.2 below. A key input in the argument is a result of Neubauer and Saltman [10], and we begin by recasting it form that is convenient for our application.

Proposition 5.1. *Let k be a field and A a finitely dimensional commutative k -algebra. Assume the k -algebra A is two-generated. For any finitely generated faithful A -module M , there are inequalities*

$$\text{rank}_k A \leq \text{rank}_k M \leq \text{rank}_k \text{End}_A(M).$$

Moreover, if either inequality is an equality, then both become equalities.

There is an analogue of this result, due to Schur and Jacobson, for arbitrary A but that involves a term that is quadratic in $\text{rank}_k M$; see [4] for an elegant proof of this result. However, this does not seem to help for applications to the conjecture of Huneke and Wiegand.

Proof. The hypothesis that A is two-generated and M is a faithful A -module is tantamount to saying that there are commuting k -linear operators $x, y: M \rightarrow M$ such that $A \cong k[x, y]$, the k -subalgebra of $\text{End}_k(M)$ generated by x, y . Since $\text{End}_A(M)$ is the k -subalgebra of $\text{End}_k(M)$ consisting of operators that commute with both x and y , it remains to apply [10, Theorem 1.1, Lemmas 1.2 & 1.3]. $\square$

Here is the avowed result about rigid ideals. We write $\text{emb dim } A$ to denote the embedding dimension of a local ring A .

Proposition 5.2. *Let $(R, \mathfrak{m}, k)$ be a Gorenstein local domain of Krull dimension one. Assume $\text{emb dim } R \leq 3$ and that there exists an element u in $\mathfrak{m} \setminus \mathfrak{m}^2$ such that the artinian ring R/Ru contains k .*

Every rigid ideal in R is principal; equivalently, free.

Proof. The result is well-known when $\text{emb dim } R \leq 2$; see [8, Theorem 3.1]. In the rest of the proof we assume $\text{emb dim } R = 3$.

Let I be a rigid ideal in R . Set $S := R/Ru$ and $M := I/Iu$. Our hypothesis on means that S is a local k -algebra, of embedding dimension two.

To verify I is free, it suffices to verify that the S -module M is free; see, for instance, [2, ??]. To that end, since the ring S is Gorenstein, it suffices to verify that M is a faithful S -module, with $\text{rank}_k S = \text{rank}_S M$.

Applying $\text{Hom}_R(I, -)$ to the exact sequence of R -modules

$$0 \longrightarrow I \xrightarrow{u} I \longrightarrow M \longrightarrow 0$$

yields the exact sequence of R -modules

$$0 \longrightarrow \text{End}_R(I) \xrightarrow{u} \text{End}_R(I) \longrightarrow \text{Hom}_R(I, M) \longrightarrow \text{Ext}_R^1(I, I) = 0$$

The equality holds because I is rigid. Observing that $\text{Hom}_R(I, M) \cong \text{End}_S(M)$, one deduces from the sequence above that

$$\text{End}_S(M) \cong \frac{\text{End}_R(I)}{\text{End}_R(I)u}$$

as S -modules. This justifies the first equality below

$$(5.1) \quad \text{rank}_k \text{End}_S(M) = \text{rank}_k \left(\frac{\text{End}_R(I)}{\text{End}_R(I)u} \right) = \text{rank}_k S = \text{rank}_k M.$$

The second and third equalities holds because $\text{End}_R(I)$ and I are both R -modules of rank one, and R is a domain of Krull dimension one; see, for instance, [2].

Let $\overline{S}$ be the image of S in $\text{End}_S(M)$ under the natural map $S \rightarrow \text{End}_S(M)$. Then $\overline{S}$ is also a commutative local k -algebra, generated by two elements. Thus Proposition 5.1 yields inequalities

$$\text{rank}_k \overline{S} \leq \text{rank}_k M \leq \text{rank}_k \text{End}_S(M),$$

and also that if either inequality is an equality, then both are. Given (5.1) it follows that there are equalities

$$\text{rank}_k S = \text{rank}_k \overline{S} = \text{rank}_k \text{End}_S(M).$$

We deduce that the natural map is an isomorphism: $S \xrightarrow{\cong} \text{End}_S(M)$, in particular, that M is a faithful S -module. $\square$

Remark 5.3. A Gorenstein ring R satisfying $\text{emb dim } R - \dim R \leq 2$ is complete intersection. Thus the ring R in Proposition 5.2 is a complete intersection. It remains an open question whether the conjecture of Huneke and Wiegand holds when R is, in addition, a complete intersection.

REFERENCES

1. Maurice Auslander and Idun Reiten, *On a generalized version of the Nakayama conjecture*, Proc. Amer. Math. Soc. **52** (1975), 69–74. MR 389977
2. W. Bruns and J. Herzog, *Cohen–Macaulay rings*, Cambridge Studies in Advanced Mathematics, vol. 39, Cambridge University Press, 1998, Revised edition.
3. *Macaulay2 script for verification*, 2026, available at <https://github.com/larswinther/homepage/tree/main/download/hwc.m2>.
4. R. C. Cowsik, *A short note on the Schur–Jacobson theorem*, Proc. Amer. Math. Soc. **118** (1993), no. 2, 675–676. MR 1139475
5. David Eisenbud and Sorin Popescu, *The projective geometry of the Gale transform*, J. Algebra **230** (2000), no. 1, 127–173. MR 1774761
6. Daniel R. Grayson and Michael E. Stillman, *Macaulay2, a software system for research in algebraic geometry*, Available at <https://faculty.math.illinois.edu/Macaulay2/>.
7. Craig Huneke, Srikanth B. Iyengar, and Roger Wiegand, *Rigid ideals in Gorenstein rings of dimension one*, Acta Math. Vietnam. **44** (2019), no. 1, 31–49. MR 3935288
8. Craig Huneke and Roger Wiegand, *Tensor products of modules and the rigidity of Tor*, Math. Ann. **299** (1994), no. 3, 449–476. MR 1282227
9. T. Y. Lam, *Lectures on modules and rings*, Graduate Texts in Mathematics, vol. 189, Springer-Verlag, Berlin/New York, 1999.
10. Michael G. Neubauer and David J. Saltman, *Two-generated commutative subalgebras of $M_n(F)$* , J. Algebra **164** (1994), no. 2, 545–562. MR 1271255
11. Son Pham, *Request for verification: An ai-assisted candidate counterexample to the Huneke–Wiegand Conjecture*, 2026, Verification repository and manuscript available at <https://github.com/sonpham-org/huneke-wiegand-candidate-verification>.
12. Jan Stevens, *Non-smoothable curve singularities*, J. Singul. **29** (2026), 100–122. MR MR5081219

TEXAS TECH UNIVERSITY, LUBBOCK, TX 79409, U.S.A.

Email address: `lars.w.christensen@ttu.edu`

URL: `https://larswinther.github.io/homepage/`

XTX MARKETS, R7 14-18 HANDYSIDE STREET LONDON, N1C 4DN, U.K.

Email address: `alexander.gerko@xtxmarkets.com`

UNIVERSITY OF UTAH, SALT LAKE CITY, UT 84112, U.S.A.

Email address: `srikanth.b.iyengar@utah.edu`